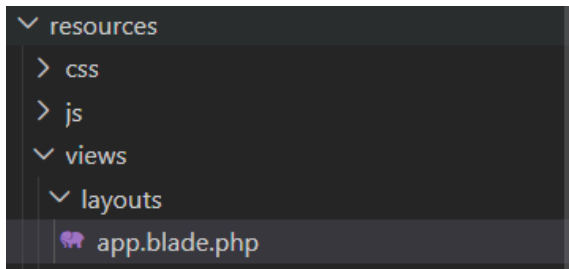


Nama : Chelsea Brilliant Syah Putra
NIM : 242410101079
Kelas : A

LAPORAN PEMROGRAMAN WEBSITE

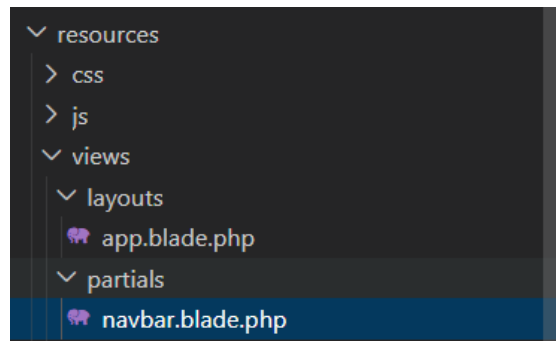
Aktivitas Mandiri 3

1. Buat layout master: resources/views/layouts/app.blade.php dengan navbar, main content area (@yield('content')), dan footer



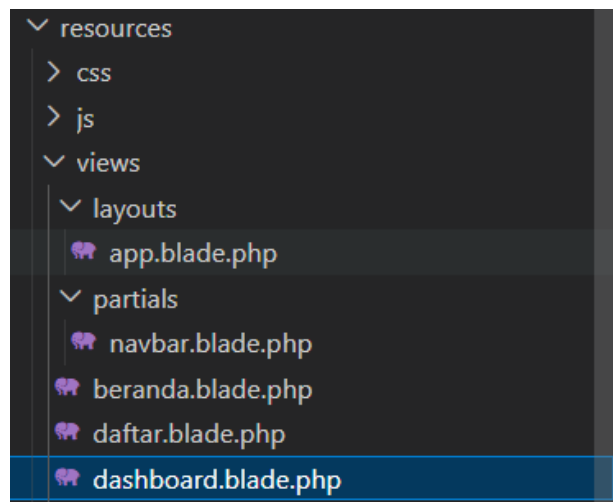
```
resources > views > layouts > app.blade.php > html > body
1 <!DOCTYPE html>
2 <html lang="id">
3   <head>
4     <meta charset="UTF-8" />
5     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6     <title>@yield('title', 'Sistem Informasi Perpustakaan')</title>
7     <link rel="preconnect" href="https://fonts.googleapis.com">
8     <link href="https://fonts.googleapis.com/css2?family=Lora:wght@500;600;700&family=Plus+Jakarta+Sans" />
9     <link rel="stylesheet" href="{{ asset('css/style.css') }}" />
10    @stack('styles')
11  </head>
12  <body>
13
14    {{-- Navbar partial --}}
15    @include('partials.navbar')
16
17    {{-- Main Content Area --}}
18    @yield('content')
19
20    {{-- Footer --}}
21    <footer>
22      <p>© 2026 | Sistem Informasi Perpustakaan | Chelsea Brilliant</p>
23    </footer>
```

2. Buat partial: resources/views/partials/navbar.blade.php dan sisipkan dengan @include di layout

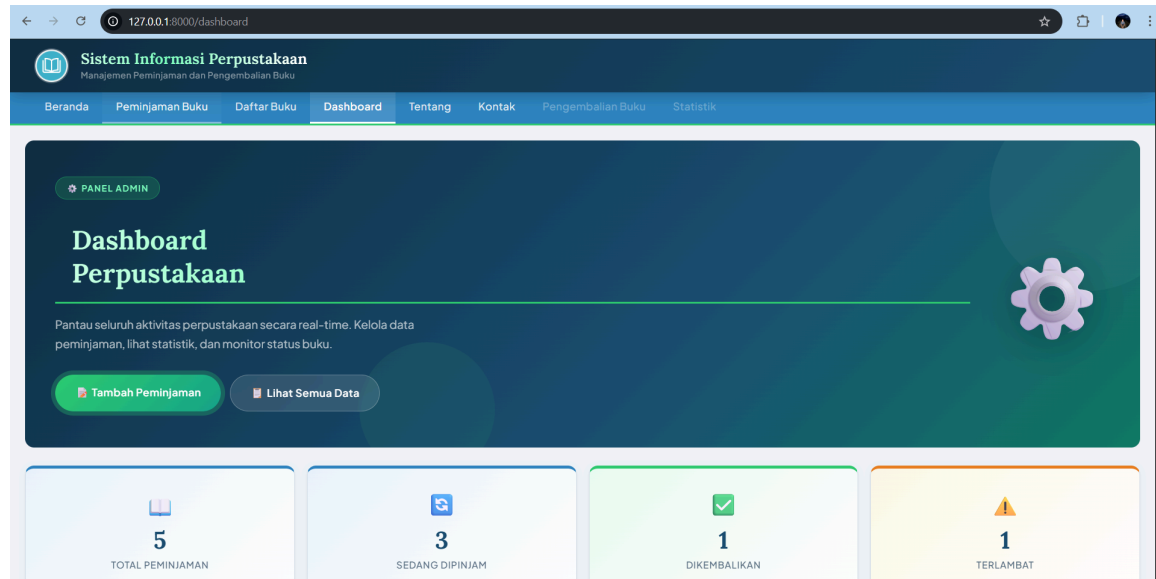


```
resources > views > partials > navbar.blade.php > ...
1 <header>
2 
4 <h1>Sistem Informasi Perpustakaan</h1>
5 <p>Manajemen Peminjaman dan Pengembalian Buku</p>
6 </div>
7 </header>
8
9 <hr />
10
11 <nav>
12 <ul>
13 <li><a href="{{ route('beranda') }}" class="{{ request()->routeIs('beranda') ? 'nav-active' : ''
14 <li><a href="{{ route('peminjaman') }}" class="{{ request()->routeIs('peminjaman') ? 'nav-active' : ''
15 <li><a href="{{ route('daftar') }}" class="{{ request()->routeIs('daftar') ? 'nav-active' : ''
16 <li><a href="{{ route('dashboard') }}" class="{{ request()->routeIs('dashboard') ? 'nav-active' : ''
17 <li><a href="{{ route('tentang') }}" class="{{ request()->routeIs('tentang') ? 'nav-active' : ''
18 <li><a class="nav-disabled">Pengembalian Buku</a></li>
19 <li><a class="nav-disabled">Statistik</a></li>
20 </ul>
21 </nav>
```

3. Buat halaman dashboard: resources/views/dashboard.blade.php yang @extends('layouts.app')

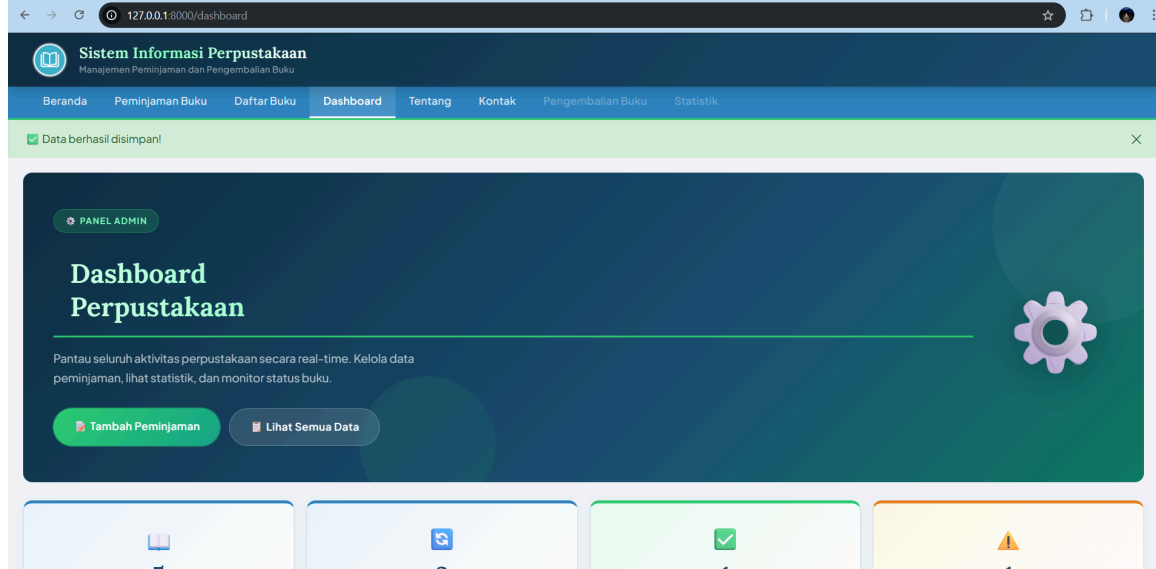


```
resources > views > 📁 dashboard.blade.php > ...
1  @extends('layouts.app')
2
3  @section('title', 'Dashboard – Sistem Informasi Perpustakaan')
4
5  @section('content')
6
7      <section id="dashboard" class="beranda">
8
9          <div class="beranda-hero" style="background: linear-gradient(135deg, #0f2e42 0%, #1b3d5b 100%);">
10             <div class="beranda-hero-content">
11                 <div class="beranda-badge">⚙️ Panel Admin</div>
12                 <h2>Dashboard<br/>Perpustakaan</h2>
13                 <p>Pantau seluruh aktivitas perpustakaan secara real-time. Kelola data peminjaman, lihat statistik, dan monitor status buku.</p>
14                 <div class="beranda-cta">
15                     <a href="{{ route('peminjaman') }}" class="cta-btn cta-primary">📖 Tambah Peminjaman</a>
16                     <a href="{{ route('daftar') }}" class="cta-btn cta-secondary">📋 Lihat Semua Data</a>
17                 </div>
18             </div>
19             <div class="beranda-hero-art">
20                 <div class="hero-book-icon">📖</div>
21             </div>
22         </div>
23     </section>
```



4. Tampilkan flash session message di layout: `@if(session('success')) ... @endif`

```
resources > views > layouts > app.blade.php > html > body
2 <html lang="id">
3 <head>
4
5 <!-- Font Family -->
6 <link href="https://fonts.googleapis.com/css2?family=Lora:wght@500;600;700&family=Plus+Jakarta+Sans" rel="stylesheet" />
7
8 <link href="https://fonts.googleapis.com/css2?family=Lora:wght@500;600;700&family=Plus+Jakarta+Sans" rel="stylesheet" />
9 <link rel="stylesheet" href="{{ asset('css/style.css') }}" />
10 @stack('styles')
11 </head>
12 <body>
13
14 {{-- Flash Session Message --}}
15 @if(session('success'))
16 <div class="alert alert-success" id="flash-message" style="background: #d4edda;color: #155724;border: 1px solid #c3e6cb;">
17     <div>{{ session('success') }}</div>
18     <button onclick="this.parentElement.style.display='none'" style="margin-left:auto;background:none;border:none;"></button>
19 </div>
20 @endif
21
22 @if(session('error'))
23 <div class="alert alert-error" id="flash-message" style="background: #f8d7da;color: #721c24;border: 1px solid #f5c6cb;">
24     <div>{{ session('error') }}</div>
25     <button onclick="this.parentElement.style.display='none'" style="margin-left:auto;background:none;border:none;"></button>
26 </div>
27 @endif
```

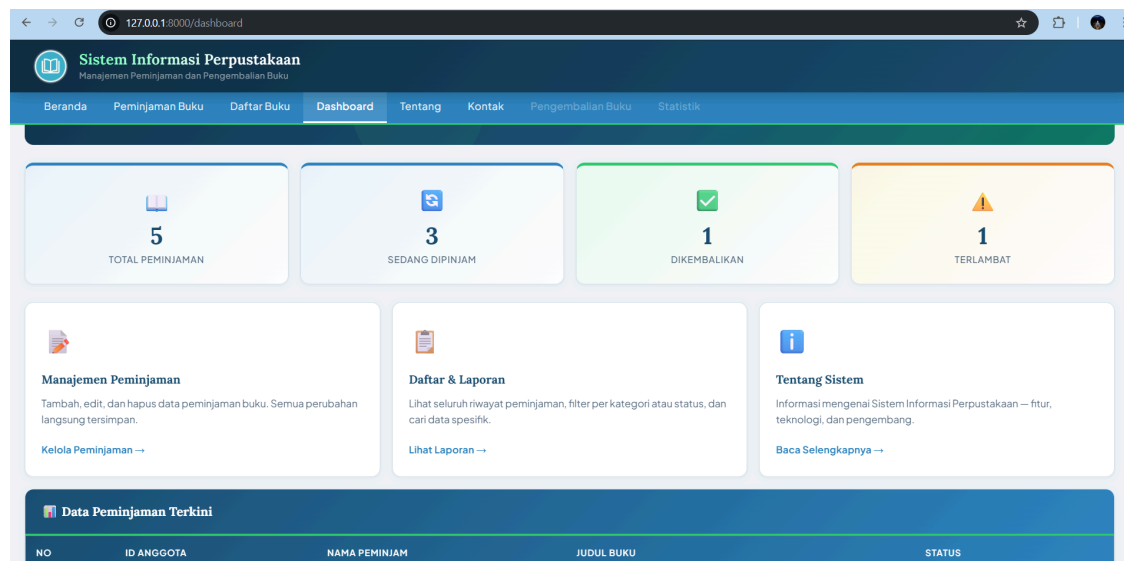


Untuk mencoba flash message berhasil atau tidak saya menggunakan `Route::get('/test-flash', function)`, dan hasilnya berhasil untuk menampilkan flash message.

5. Buat komponen: `resources/views/components/stat-card.blade.php` (kartu statistik dengan props: judul, nilai, ikon, warna)

```
views
components
stat-card.blade.php
```

```
resources > views > components > stat-card.blade.php > div.bstat-card
1  @props([
2      'judul' => 'Label',
3      'nilai' => '0',
4      'ikon' => '📊',
5      'warna' => 'default'
6  ])
7
8  @php
9      $styles = match($warna) {
10         'success' => 'border-top: 4px solid #2ecc71; background: linear-gradient(135deg, #eafaf1, #ffffff);',
11         'warning' => 'border-top: 4px solid #e67e22; background: linear-gradient(135deg, #fef9e7, #ffffff);',
12         'danger' => 'border-top: 4px solid #e74c3c; background: linear-gradient(135deg, #fdedec, #ffffff);',
13         default => 'border-top: 4px solid #2e86c1; background: linear-gradient(135deg, #eaf4fb, #ffffff);',
14     };
15  @endphp
16
17  <div class="bstat-card" style="{{ $styles }}">
18      <div class="bstat-icon">{{ $ikon }}</div>
19      <div class="bstat-num">{{ $nilai }}</div>
20      <div class="bstat-label">{{ $judul }}</div>
21  </div>
```



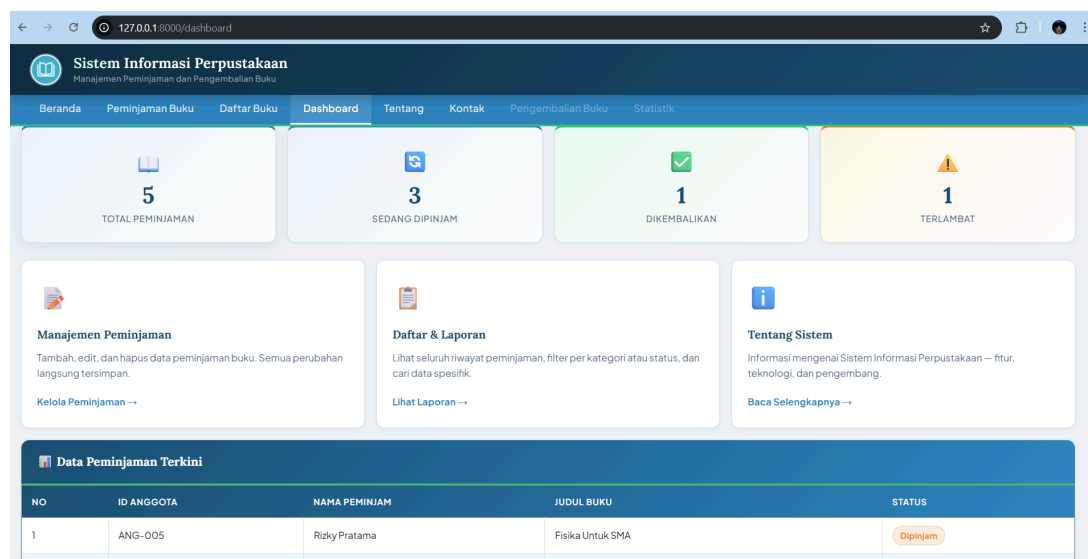
- Gunakan @forelse di dashboard untuk loop array data dummy (jumlah mahasiswa, mata kuliah, dll)

```
resources > views > dashboard.blade.php > ...
7  <section id="dashboard" class="beranda">
33  <div class="beranda-fitur">
34      @forelse($menuAdmin as $menu)
35          <div class="fitur-card">
36              <div class="fitur-icon">{{ $menu['ikon'] }}</div>
37              <h3>{{ $menu['judul'] }}</h3>
38              <p>{{ $menu['deskripsi'] }}</p>
39              <a href="{{ route($menu['route']) }}" class="fitur-link">{{ $menu['label'] }} -></a>
40          </div>
41      @empty
42          <p>Menu tidak tersedia.</p>
43      @endforelse
44  </div>
```

```

app > Http > Controllers > DashboardController.php > DashboardController > index
7 class DashboardController extends Controller
9     public function index()
19         // Data dummy menu admin - digunakan oleh @forelse
20         $menuAdmin = [
21             [
22                 'ikon' => '📖',
23                 'judul' => 'Manajemen Peminjaman',
24                 'deskripsi' => 'Tambah, edit, dan hapus data peminjaman buku. Semua perubahan langsung tersimpan.',
25                 'route' => 'peminjaman',
26                 'label' => 'Kelola Peminjaman',
27             ],
28             [
29                 'ikon' => '📋',
30                 'judul' => 'Daftar & Laporan',
31                 'deskripsi' => 'Lihat seluruh riwayat peminjaman, filter per kategori atau status, dan cari data spesifik.',
32                 'route' => 'daftar',
33                 'label' => 'Lihat Laporan',
34             ],
35             [
36                 'ikon' => 'ℹ️',
37                 'judul' => 'Tentang Sistem',
38                 'deskripsi' => 'Informasi mengenai Sistem Informasi Perpustakaan – fitur, teknologi, dan pengembangan.',
39                 'route' => 'tentang',

```



DashboardController diupdate untuk mengirimkan array data dummy \$statistik dan \$menuAdmin ke view. Di dashboard.blade.php, @forelse digunakan untuk loop array \$menuAdmin dan menampilkan kartu fitur admin secara dinamis.

7. Buat halaman 'Tentang' dan 'Kontak' yang @extends layout yang sama

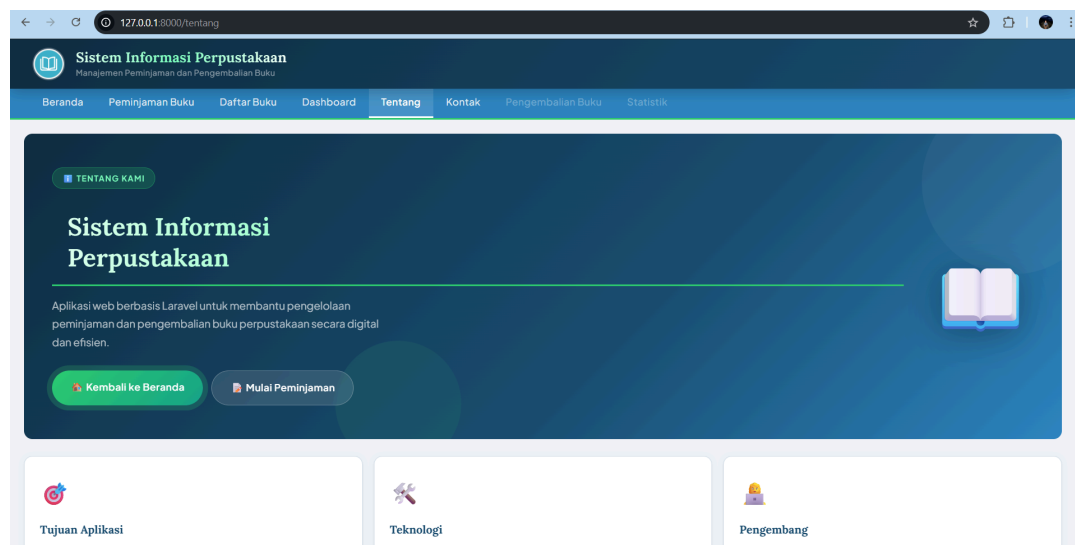
```

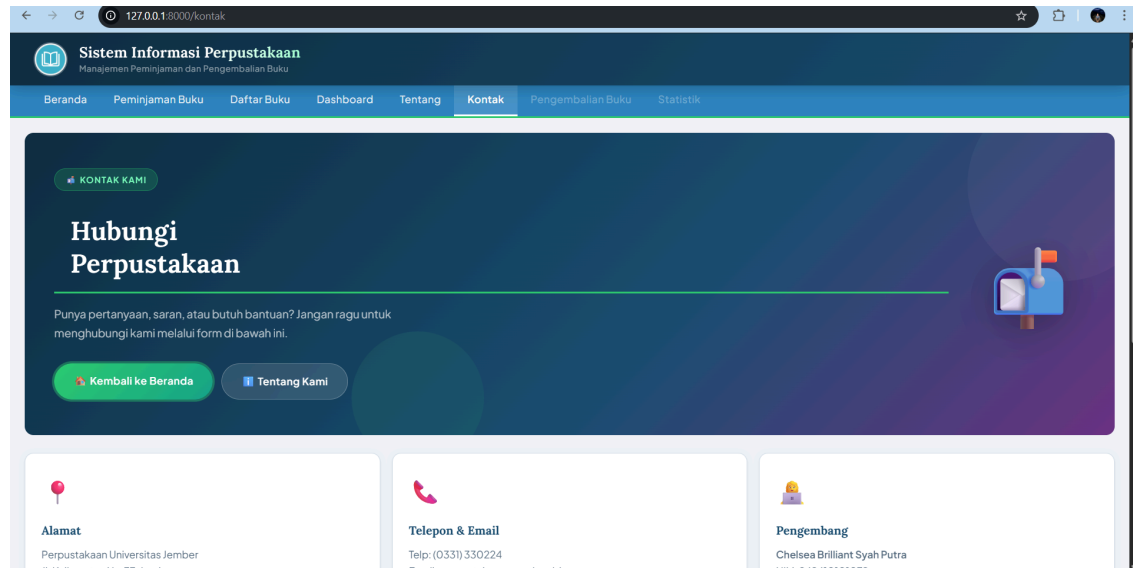
🐘 kontak.blade.php
🐘 layout.blade.php
🐘 peminjaman.blade.php
🐘 tentang.blade.php

```

```
tentang.blade.php X
perpustakaan-pweb > resources > views > tentang.blade.php > ...
1 @extends('layouts.app')
2
3 @section('title', 'Tentang – Sistem Informasi Perpustakaan')
4
5 @section('content')
6
7     <section id="tentang" class="beranda">
8
9         <div class="beranda-hero" style="background: linear-gradient(135deg, #0f2e42 0%, #1b3d5b 60%, #2d5a78 100%);">
10             <div class="beranda-hero-content">
11                 <div class="beranda-badge">📖 Tentang Kami</div>
12                 <h2>Sistem Informasi<br/>Perpustakaan</h2>
13                 <p>Aplikasi web berbasis Laravel untuk membantu pengelolaan peminjaman dan pengembalian buku perpustakaan secara digital dan efisien.</p>
14                 <div class="beranda-cta">
15                     <a href="{{ route('beranda') }}" class="cta-btn cta-primary">🏠 Kembali ke Beranda</a>
16                     <a href="{{ route('peminjaman') }}" class="cta-btn cta-secondary">📖 Mulai Peminjaman</a>
17                 </div>
18             </div>
19             <div class="beranda-hero-art">
20                 <div class="hero-book-icon">📖</div>
21             </div>
22         </div>
23     </section>
24 </div>
```

```
resources > views > kontak.blade.php > ...
1 @extends('layouts.app')
2
3 @section('title', 'Kontak – Sistem Informasi Perpustakaan')
4
5 @section('content')
6
7     <section id="kontak" class="beranda">
8
9         <div class="beranda-hero" style="background: linear-gradient(135deg, #0f2e42 0%, #1b3d5b 60%, #2d5a78 100%);">
10             <div class="beranda-hero-content">
11                 <div class="beranda-badge">📞 Kontak Kami</div>
12                 <h2>Hubungi<br/>Perpustakaan</h2>
13                 <p>Punya pertanyaan, saran, atau butuh bantuan? Jangan ragu untuk menghubungi kami melalui form di bawah ini.</p>
14                 <div class="beranda-cta">
15                     <a href="{{ route('beranda') }}" class="cta-btn cta-primary">🏠 Kembali ke Beranda</a>
16                     <a href="{{ route('tentang') }}" class="cta-btn cta-secondary">📖 Tentang Kami</a>
17                 </div>
18             </div>
19             <div class="beranda-hero-art">
20                 <div class="hero-book-icon">📖</div>
21             </div>
22         </div>
23     </section>
24 </div>
```





Dibuat dua halaman yang keduanya menggunakan `@extends` layout yang sama (`layouts.app`). Halaman Tentang sudah ada sebelumnya, kemudian ditambahkan halaman Kontak baru beserta `KontakController` dan route `/kontak`.

- Gunakan `@push('scripts')` dan `@stack('scripts')` untuk menambahkan JavaScript spesifik per halaman

resources > views > layouts > app.blade.php > html > body

```
2    <html lang="id">
12    <body>
27    @endif
28
29    {{-- Navbar partial --}}
30    @include('partials.navbar')
31
32    {{-- Main Content Area --}}
33    @yield('content')
34
35    {{-- Footer --}}
36    <footer>
37    <p>© 2026 | Sistem Informasi Perpustakaan | Chelsea Brilliant</p>
38    </footer>
39
40    <script src="{{ asset('js/script.js') }}"></script>
41    @stack('scripts')
42    </body>
43    </html>
```

resources > views > beranda.blade.php > script

```
7    <section id="beranda" class="beranda">
46    <div class="beranda-fitur">
59    <div class="fitur-card">
62
63    <p>Cari data peminjaman berdasarkan nama, judul, kode buku,
64    <a href="{{ route('daftar') }}" class="fitur-link">Cari Dat
65    </div>
66    </div>
67    </section>
68    @endsection
69
70    @push('scripts')
71    <script>
72    document.addEventListener('DOMContentLoaded', () => {
73    console.log('Halaman Beranda loaded');
74    });
75    </script>
76    @endpush
77
```

```
resources > views > 🐛 dashboard.blade.php > script
71 @endsection
72
73 @push('scripts')
74 <script>
75 document.addEventListener('DOMContentLoaded', () => {
76     const tbody = document.getElementById('dashboard-tabel');
77     if (!tbody) return;
78
79     const data = JSON.parse(localStorage.getItem('perpustakaan_data')) || [];
80     const statusBadge = (status) => {
81         const map = { 'Dikembalikan': 'badge-success', 'Dipinjam': 'badge-warning', 'Terlambat': 'badge-danger';
82         return `<span class="badge ${map[status] || ''}">${status}</span>`;
83     };
84
85     if (data.length === 0) {
86         tbody.innerHTML = `<tr><td colspan="5" style="text-align:center;padding:24px;color:#5d7285;">Belum ada
87         return;
88     }
89 }
```

```
resources > views > 🐛 kontak.blade.php > script
7 <section id="kontak" class="beranda"
44 <div style="background: #f1f3f4; border: 1px solid #ccc; padding: 10px; margin-top: 10px;">
76 </div>
77
78 </section>
79
80 @endsection
81
82 @push('scripts')
83 <script>
84 console.log('Halaman Kontak loaded');
85 </script>
86 @endpush
```

```
resources > views > 🐛 layout.blade.php > ...
2 <html lang="id">
11 <body>
23 <nav>
24 <ul>
32 </ul>
33 </nav>
34
35 <hr />
36
37 @yield('content')
38
39 <footer>
40 <p>© 2026 | Sistem Informasi Perpustakaan | Chelsea Brilliant</p>
41 </footer>
42
43 <script src="{{ asset('js/script.js') }}"></script>
44 @stack('scripts')
45 </body>
46 </html>
```

```
resources > views > tentang.blade.php > script
/
<section id= 'tentang' class= 'beranda' >
94     </section>
95
96 @endsection
97
98 @push('scripts')
99 <script>
100 document.addEventListener('DOMContentLoaded', () => {
101     console.log('Halaman Tentang loaded');
102 });
103 </script>
104 @endpush
```

Setiap halaman yang membutuhkan JavaScript khusus menggunakan `@push('scripts')` di bawah `@endsection`. Di layout master `layouts/app.blade.php` terdapat `@stack('scripts')` yang akan menerima dan merender semua script yang di-push dari halaman manapun.

Aktivitas Mandiri 4

1. Buat Model dan Migration sekaligus: `php artisan make:model Mahasiswa -m`

```
PS C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb> php artisan make:model Peminjaman -m

INFO Model [C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb\app\Models\Peminjaman.php] created successfully.

INFO Migration [C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb\database\Migrations\2026_05_03_123013_create_peminjaman_table.php] created successfully.

PS C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb> php artisan make:model Buku -m

INFO Model [C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb\app\Models\Buku.php] created successfully.
```

2. Edit file migration: tambahkan kolom `nim`, `nama`, `email`, `jurusan (enum)`, `ipk (decimal)`, `semester`, `aktif (boolean)`, `foto (nullable)`

database > migrations > 2026_05_03_123013_create_peminjaman_table.php > class > down

```
1  <?php
2
3  use Illuminate\Database\Migrations\Migration;
4  use Illuminate\Database\Schema\Blueprint;
5  use Illuminate\Support\Facades\Schema;
6
7  return new class extends Migration
8  {
9      public function up(): void
10     {
11         Schema::create('peminjaman', function (Blueprint $table) {
12             $table->id();
13             $table->string('id_anggota');
14             $table->string('nama_peminjam');
15             $table->string('judul_buku');
16             $table->string('kode_buku')->nullable();
17             $table->enum('status', ['Dipinjam', 'Dikembalikan', 'Terlambat'])->default('Dipinjam');
18             $table->date('tanggal_pinjam');
19             $table->date('tanggal_kembali')->nullable();
20             $table->boolean('aktif')->default(true);
21             $table->text('keterangan')->nullable();
22             $table->timestamps();
23         });
24     }
25
26     public function down(): void
27     {
28         Schema::dropIfExists('peminjaman');
29     }
30 };
31
```

```

database > migrations > 2026_05_03_123026_create_bukus_table.php > class > down
1  <?php
2
3  use Illuminate\Database\Migrations\Migration;
4  use Illuminate\Database\Schema\Blueprint;
5  use Illuminate\Support\Facades\Schema;
6
7  return new class extends Migration
8  {
9      public function up(): void
10     {
11         Schema::create('buku', function (Blueprint $table) {
12             $table->id();
13             $table->string('kode_buku')->unique();
14             $table->string('judul');
15             $table->string('pengarang');
16             $table->enum('kategori', ['Fiksi', 'Non-Fiksi', 'Sains', 'Teknologi', 'Sejarah', 'Agama']);
17             $table->integer('tahun_terbit');
18             $table->integer('stok')->default(1);
19             $table->boolean('tersedia')->default(true);
20             $table->string('foto')->nullable();
21             $table->timestamps();
22         });
23     }
24
25     public function down(): void
26     {
27         Schema::dropIfExists('buku');
28     }
29 };
30

```

3. Jalankan migration: php artisan migrate — cek di phpMyAdmin bahwa tabel sudah terbuat

PS C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb> php artisan migrate

INFO Running migrations.

```

0001_01_01_000000_create_users_table ..... 34.67ms DONE
0001_01_01_000001_create_cache_table ..... 16.40ms DONE
0001_01_01_000002_create_jobs_table ..... 18.61ms DONE
2026_05_03_123013_create_peminjaman_table ..... 4.02ms DONE
2026_05_03_123026_create_bukus_table ..... 7.81ms DONE

```

Server: 127.0.0.1 » Database: perpustakaan_pweb » Table: peminjaman

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

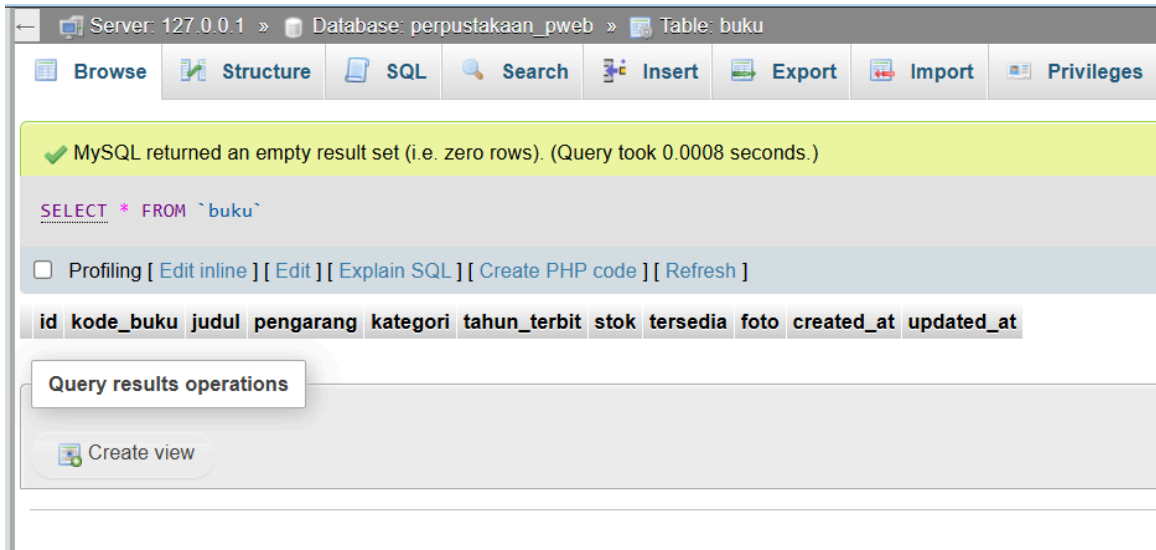
SELECT * FROM `peminjaman`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

id	id_anggota	nama_peminjam	judul_buku	kode_buku	status	tanggal_pinjam	tanggal_kembali	aktif	keterangan	created_at	updated_at
----	------------	---------------	------------	-----------	--------	----------------	-----------------	-------	------------	------------	------------

Query results operations

Create view



4. Edit Model Mahasiswa: isi \$fillable, tambahkan \$casts, dan buat local scope scopeAktif()

```

app > Models > Peminjaman.php > Peminjaman > $table
1  <?php
2
3  namespace App\Models;
4
5  use Illuminate\Database\Eloquent\Model;
6  use Illuminate\Database\Eloquent\Factories\HasFactory;
7
8  class Peminjaman extends Model
9  {
10     use HasFactory;
11
12     protected $table = 'peminjaman';
13
14     protected $fillable = [
15         'id_anggota',
16         'nama_peminjam',
17         'judul_buku',
18         'kode_buku',
19         'status',
20         'tanggal_pinjam',
21         'tanggal_kembali',
22         'aktif',
23         'keterangan',
24     ];
25
26     protected $casts = [
27         'tanggal_pinjam' => 'date',
28         'tanggal_kembali' => 'date',
29         'aktif' => 'boolean',
30     ];
31
32     // Local Scope: hanya peminjaman aktif
33     public function scopeAktif($query)
34     {
35         return $query->where('aktif', true);
36     }

```

```

38 // Local Scope: filter berdasarkan status
39 public function scopeStatus($query, $status)
40 {
41     return $query->where('status', $status);
42 }
43
44 // Relasi belongsToMany ke Buku
45 public function buku()
46 {
47     return $this->belongsToMany(Buku::class, 'peminjaman_buku', 'peminjaman_id', 'buku_id');
48 }
49
50

```

```

app > Models > Buku.php > Buku > peminjaman
1  <?php
2
3  namespace App\Models;
4
5  use Illuminate\Database\Eloquent\Model;
6  use Illuminate\Database\Eloquent\Factories\HasFactory;
7
8  class Buku extends Model
9  {
10     use HasFactory;
11
12     protected $table = 'buku';
13
14     protected $fillable = [
15         'kode_buku',
16         'judul',
17         'pengarang',
18         'kategori',
19         'tahun_terbit',
20         'stok',
21         'tersedia',
22         'foto',
23     ];
24
25     protected $casts = [
26         'tersedia' => 'boolean',
27         'tahun_terbit' => 'integer',
28         'stok' => 'integer',
29     ];
30
31     // Local Scope: buku yang tersedia
32     public function scopeTersedia($query)
33     {
34         return $query->where('tersedia', true);
35     }
36
37     // Relasi belongsToMany ke Peminjaman
38     public function peminjaman()
39     {
40         return $this->belongsToMany(Peminjaman::class, 'peminjaman_buku', 'buku_id', 'peminjaman_id');
41     }
42 }
43

```

5. Buat seeder: php artisan make:seeder MahasiswaSeeder — isi dengan 5-10 data dummy

```
PS C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb> php artisan make:seeder PeminjamanSeeder

INFO Seeder [C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb\database\seeders\PeminjamanSeeder.php] created successfully.

PS C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb>
```

```
PS C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb> php artisan make:seeder BukuSeeder

INFO Seeder [C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb\database\seeders\BukuSeeder.php] created successfully.
```

```
database > seeders > PeminjamanSeeder.php > ...
1  <?php
2
3  namespace Database\Seeders;
4
5  use Illuminate\Database\Seeder;
6  use App\Models\Peminjaman;
7
8  class PeminjamanSeeder extends Seeder
9  {
10     public function run(): void
11     {
12         $data = [
13             [
14                 'id_anggota' => 'ANG-001',
15                 'nama_peminjam' => 'Andi Pratama',
16                 'judul_buku' => 'Pemrograman Laravel 10',
17                 'kode_buku' => 'BK-001',
18                 'status' => 'Dipinjam',
19                 'tanggal_pinjam' => '2025-04-01',
20                 'tanggal_kembali' => null,
21                 'aktif' => true,
22                 'keterangan' => null,
23             ],
24             [
25                 'id_anggota' => 'ANG-002',
26                 'nama_peminjam' => 'Budi Santoso',
27                 'judul_buku' => 'Clean Code: A Handbook',
28                 'kode_buku' => 'BK-002',
29                 'status' => 'Dikembalikan',
30                 'tanggal_pinjam' => '2025-03-15',
31                 'tanggal_kembali' => '2025-03-28',
32                 'aktif' => true,
33                 'keterangan' => 'Dikembalikan tepat waktu',
34             ],
35             [
36                 'id_anggota' => 'ANG-003',
37                 'nama_peminjam' => 'Citra Dewi',
```



```

database > seeders > BukuSeeder.php > ...
1  <?php
2
3  namespace Database\Seeders;
4
5  use Illuminate\Database\Seeder;
6  use App\Models\Buku;
7
8  class BukuSeeder extends Seeder
9  {
10     public function run(): void
11     {
12         $data = [
13             ['kode_buku' => 'BK-001', 'judul' => 'Pemrograman Laravel 10', 'pengarang' => 'Imam Syahrini'],
14             ['kode_buku' => 'BK-002', 'judul' => 'Clean Code: A Handbook', 'pengarang' => 'Robert C. Martin'],
15             ['kode_buku' => 'BK-003', 'judul' => 'Fisika Untuk SMA', 'pengarang' => 'Marthen Kanginan'],
16             ['kode_buku' => 'BK-004', 'judul' => 'Algoritma dan Pemrograman', 'pengarang' => 'Rinaldi Munir'],
17             ['kode_buku' => 'BK-005', 'judul' => 'Basis Data dan SQL', 'pengarang' => 'Fathansyah'],
18             ['kode_buku' => 'BK-006', 'judul' => 'Jaringan Komputer Dasar', 'pengarang' => 'Andrew Tanenbaum'],
19             ['kode_buku' => 'BK-007', 'judul' => 'Kalkulus Lanjut', 'pengarang' => 'James Stewart'],
20         ];
21
22         foreach ($data as $item) {
23             Buku::create($item);
24         }
25     }
26 }
27

```

6. Daftarkan seeder di DatabaseSeeder.php dan jalankan: php artisan db:seed

```

database > seeders > DatabaseSeeder.php > DatabaseSeeder
1  <?php
2
3  namespace Database\Seeders;
4
5  use Illuminate\Database\Seeder;
6
7  class DatabaseSeeder extends Seeder
8  {
9
10     public function run(): void
11     {
12         $this->call([
13             BukuSeeder::class,
14             PeminjamanSeeder::class,
15         ]);
16     }
17 }

```

```

PS C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb> php artisan db:seed

INFO Seeding database.

Database\Seeders\BukuSeeder ..... RUNNING
Database\Seeders\BukuSeeder ..... 29 ms DONE

Database\Seeders\PeminjamanSeeder ..... RUNNING
Database\Seeders\PeminjamanSeeder ..... 8 ms DONE

```

7. Buka php artisan tinker dan coba query: Mahasiswa::all(), Mahasiswa::where('jurusan','SI')->get(), Mahasiswa::avg('ipk')

```
PS C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb> php artisan tinker
Psy Shell v0.12.22 (PHP 8.3.30 - cli) by Justin Hileman
New PHP manual is available (latest: 3.0.5). Update with `doc --update-manual`
```

```
> Peminjaman::all()
```

```
[!] Aliasing 'Peminjaman' to 'App\Models\Peminjaman' for this Tinker session.
```

```
= Illuminate\Database\Eloquent\Collection {#7800}
```

```
all: [
    App\Models\Peminjaman {#7807
        id: 1,
        id_anggota: "ANG-001",
        nama_peminjam: "Andi Pratama",
        judul_buku: "Pemrograman Laravel 10",
        kode_buku: "BK-001",
        status: "Dipinjam",
        tanggal_pinjam: "2025-04-01",
        tanggal_kembali: null,
        aktif: 1,
        keterangan: null,
        created_at: "2026-05-03 13:16:52",
        updated_at: "2026-05-03 13:16:52",
    },
    App\Models\Peminjaman {#7808
        id: 2,
        id_anggota: "ANG-002",
        nama_peminjam: "Budi Santoso",
        judul_buku: "Clean Code: A Handbook",
        kode_buku: "BK-002",
        status: "Dikembalikan",
        tanggal_pinjam: "2025-03-15",
        tanggal_kembali: "2025-03-28",
        aktif: 1,
        keterangan: "Dikembalikan tepat waktu",
        created_at: "2026-05-03 13:16:52",
        updated_at: "2026-05-03 13:16:52",
    },
]
```

```
PS C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb> php artisan tinker
```

```
App\Models\Peminjaman {#7805}
```

```
id: 3,
id_anggota: "ANG-003",
nama_peminjam: "Citra Dewi",
judul_buku: "Fisika Untuk SMA",
kode_buku: "BK-003",
status: "Terlambat",
tanggal_pinjam: "2025-03-01",
tanggal_kembali: null,
aktif: 1,
keterangan: "Sudah melewati batas waktu",
created_at: "2026-05-03 13:16:52",
updated_at: "2026-05-03 13:16:52",
},
```

```
App\Models\Peminjaman {#7804}
```

```
id: 4,
id_anggota: "ANG-004",
nama_peminjam: "Dina Rahayu",
judul_buku: "Algoritma dan Pemrograman",
kode_buku: "BK-004",
status: "Dipinjam",
tanggal_pinjam: "2025-04-10",
tanggal_kembali: null,
aktif: 1,
keterangan: null,
created_at: "2026-05-03 13:16:52",
updated_at: "2026-05-03 13:16:52",
},
```

```
App\Models\Peminjaman {#7803}
```

```
id: 5,
id_anggota: "ANG-005",
nama_peminjam: "Eko Wijaya",
judul_buku: "Basis Data dan SQL",
kode_buku: "BK-005",
status: "Dikembalikan",
tanggal_pinjam: "2025-03-20",
```

```
> Peminjaman::where('status','Dipinjam')->get()

= Illuminate\Database\Eloquent\Collection {#7362
  all: [
    App\Models\Peminjaman {#7379
      id: 1,
      id_anggota: "ANG-001",
      nama_peminjam: "Andi Pratama",
      judul_buku: "Pemrograman Laravel 10",
      kode_buku: "BK-001",
      status: "Dipinjam",
      tanggal_pinjam: "2025-04-01",
      tanggal_kembali: null,
      aktif: 1,
      keterangan: null,
      created_at: "2026-05-03 13:16:52",
      updated_at: "2026-05-03 13:16:52",
    },
    App\Models\Peminjaman {#7812
      id: 4,
      id_anggota: "ANG-004",
      nama_peminjam: "Dina Rahayu",
      judul_buku: "Algoritma dan Pemrograman",
      kode_buku: "BK-004",
      status: "Dipinjam",
      tanggal_pinjam: "2025-04-10",
      tanggal_kembali: null,
      aktif: 1,
      keterangan: null,
      created_at: "2026-05-03 13:16:52",
      updated_at: "2026-05-03 13:16:52",
    },
    App\Models\Peminjaman {#7798
      id: 6,
      id_anggota: "ANG-006",
      nama_peminjam: "Fajar Nugroho",
```

```
> Peminjaman::aktif()->get()

= Illuminate\Database\Eloquent\Collection {#7382
  all: [
    App\Models\Peminjaman {#7780
      id: 1,
      id_anggota: "ANG-001",
      nama_peminjam: "Andi Pratama",
      judul_buku: "Pemrograman Laravel 10",
      kode_buku: "BK-001",
      status: "Dipinjam",
      tanggal_pinjam: "2025-04-01",
      tanggal_kembali: null,
      aktif: 1,
      keterangan: null,
      created_at: "2026-05-03 13:16:52",
      updated_at: "2026-05-03 13:16:52",
    },
    App\Models\Peminjaman {#7383
      id: 2,
      id_anggota: "ANG-002",
      nama_peminjam: "Budi Santoso",
      judul_buku: "Clean Code: A Handbook",
      kode_buku: "BK-002",
      status: "Dikembalikan",
      tanggal_pinjam: "2025-03-15",
      tanggal_kembali: "2025-03-28",
      aktif: 1,
      keterangan: "Dikembalikan tepat waktu",
      created_at: "2026-05-03 13:16:52",
      updated_at: "2026-05-03 13:16:52",
    },
    App\Models\Peminjaman {#7778
      id: 3,
      id_anggota: "ANG-003",
```

```
> Peminjaman::aktif()->count()

= 6

> Peminjaman::status('Terlambat')->get()

= Illuminate\Database\Eloquent\Collection {#7770
  all: [
    App\Models\Peminjaman {#7792
      id: 3,
      id_anggota: "ANG-003",
      nama_peminjam: "Citra Dewi",
      judul_buku: "Fisika Untuk SMA",
      kode_buku: "BK-003",
      status: "Terlambat",
      tanggal_pinjam: "2025-03-01",
      tanggal_kembali: null,
      aktif: 1,
      keterangan: "Sudah melewati batas waktu",
      created_at: "2026-05-03 13:16:52",
      updated_at: "2026-05-03 13:16:52",
    },
    App\Models\Peminjaman {#7814
      id: 7,
      id_anggota: "ANG-007",
      nama_peminjam: "Gita Permata",
      judul_buku: "Kalkulus Lanjut",
      kode_buku: "BK-007",
      status: "Terlambat",
      tanggal_pinjam: "2025-02-28",
      tanggal_kembali: null,
      aktif: 1,
      keterangan: "Dihubungi via email",
      created_at: "2026-05-03 13:16:52",
      updated_at: "2026-05-03 13:16:52",
    },
  ],
}
```

```

> Buku::all()->count()

[!] Aliasing 'Buku' to 'App\Models\Buku' for this Tinker session.
= 7

> Buku::tersedia()->get()

= Illuminate\Database\Eloquent\Collection {#7891
  all: [
    App\Models\Buku {#7902
      id: 1,
      kode_buku: "BK-001",
      judul: "Pemrograman Laravel 10",
      pengarang: "Imam Syahrani",
      kategori: "Teknologi",
      tahun_terbit: 2023,
      stok: 3,
      tersedia: 1,
      foto: null,
      created_at: "2026-05-03 13:16:52",
      updated_at: "2026-05-03 13:16:52",
    },
    App\Models\Buku {#7890
      id: 2,
      kode_buku: "BK-002",
      judul: "Clean Code: A Handbook",
      pengarang: "Robert C. Martin",
      kategori: "Teknologi",
      tahun_terbit: 2008,
      stok: 2,
      tersedia: 1,
      foto: null,
      created_at: "2026-05-03 13:16:52",
      updated_at: "2026-05-03 13:16:52",
    },
    App\Models\Buku {#7382

```

```

> Buku::where('kategori','Teknologi')->get()->count()

= 5

```

8. Coba buat relasi: buat tabel mata_kuliahs dan tabel pivot krs, lalu definisikan relasi belongsToMany di Model

```

database > migrations > 2026_05_03_142639_create_peminjaman_buku_table.php > class > up > Closure
1  <?php
2
3  use Illuminate\Database\Migrations\Migration;
4  use Illuminate\Database\Schema\Blueprint;
5  use Illuminate\Support\Facades\Schema;
6
7  return new class extends Migration
8  {
9      public function up(): void
10     {
11         Schema::create('peminjaman_buku', function (Blueprint $table) {
12             $table->id();
13             $table->foreignId('peminjaman_id')->constrained('peminjaman')->onDelete('cascade');
14             $table->foreignId('buku_id')->constrained('buku')->onDelete('cascade');
15             $table->timestamps();
16         });
17     }
18
19     public function down(): void
20     {
21         Schema::dropIfExists('peminjaman_buku');
22     }
23 };
24

```

```

app > Models > Peminjaman.php > Peminjaman > $table
8  class Peminjaman extends Model
9  {
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44     // Relasi belongsToMany ke Buku
45     public function buku()
46     {
47         return $this->belongsToMany(Buku::class, 'peminjaman_buku', 'peminjaman_id', 'buku_id');
48     }
49 }
50

```

```

app > Models > Buku.php > Buku > peminjaman
8  class Buku extends Model
9  {
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37     // Relasi belongsToMany ke Peminjaman
38     public function peminjaman()
39     {
40         return $this->belongsToMany(Peminjaman::class, 'peminjaman_buku', 'buku_id', 'peminjaman_id');
41     }
42 }
43

```

```
PS C:\Users\CHELSEA\OneDrive\Documents\Perpustakaan-PWEB\perpustakaan-pweb> php artisan tinker
> $p = Peminjaman::find(1);

[!] Aliasing 'Peminjaman' to 'App\Models\Peminjaman' for this Tinker session.
= App\Models\Peminjaman {#7830
  id: 1,
  id_anggota: "ANG-001",
  nama_peminjam: "Andi Pratama",
  judul_buku: "Pemrograman Laravel 10",
  kode_buku: "BK-001",
  status: "Dipinjam",
  tanggal_pinjam: "2025-04-01",
  tanggal_kembali: null,
  aktif: 1,
  keterangan: null,
  created_at: "2026-05-03 13:16:52",
  updated_at: "2026-05-03 13:16:52",
}
```

```
> $p->buku()->attach(1);

= null

> $p->buku()->get();

= Illuminate\Database\Eloquent\Collection {#8601
  all: [
    App\Models\Buku {#7356
      id: 1,
      kode_buku: "BK-001",
      judul: "Pemrograman Laravel 10",
      pengarang: "Imam Syahroni",
      kategori: "Teknologi",
      tahun_terbit: 2023,
      stok: 3,
      tersedia: 1,
      foto: null,
      created_at: "2026-05-03 13:16:52",
      updated_at: "2026-05-03 13:16:52",
      pivot: Illuminate\Database\Eloquent\Relations\Pivot {#7820
        peminjaman_id: 1,
        buku_id: 1,
      },
    },
  ],
}
```

Relasi belongsToMany antara Peminjaman dan Buku berhasil dibuat melalui tabel pivot peminjaman_buku. Terlihat dari hasil tinker bahwa Peminjaman id=1 berhasil terhubung ke Buku id=1 dengan pivot peminjaman_id: 1 dan buku_id: 1.